Sustania: A Digital Farming Ecosystem for Smart Agriculture

1. Amisha Angel Pius[1], 2. Ronn Mathew Sino[2], 3. Divya James[3],
4. Emmanuel Santhosh[4], 5. Rohith M P[5]
Computer Science and Business Systems,
Rajagiri School of Engineering & Technology, Kochi, Kerala, India
Emails: [1]amishapius@gmail.com, [2]ronnmathewsino@gmail.com, [3]divyaj@rajagiritech.edu.in,
[4]u2209020@rajagiri.edu.in, [5]u2209057@rajagiri.edu.in
[3]Associate Professor, Department of Computer Science and Business Systems

***Abstract*—We have developed an innovative Smart Irrigation System to address recent challenges in agricultural water management and to meet the growing need for sustainable farming practices, particularly for small- and medium-scale farmers. The system integrates Internet of Things (IoT)-based sensors, machine learning algorithms, and a user-friendly web interface to optimize water usage in agricultural settings. It uses soil moisture and temperature sensors along with a Random Forest classifier, ensuring precise and reliable scheduling. The platform features a modular architecture with an SQLite backend and a React-based front-end, making it a cost-effective solution. Initial testing was conducted using a DHT11 temperature sensor and a resistive soil moisture sensor; however, subsequent evaluations demonstrated that the DHT22 temperature sensor provided better performance and durability. After extensive comparison, the Random Forest classifier was selected for model training, as it proved to be the fastest and most accurate option with an accuracy of 91%. Core features of the system include real-time temperature and moisture updates and predictive capabilities based on these readings. Our system has been named "Sustania" due to its potential to serve as a valuable tool for farmers transitioning to the modern era of sustainable and innovative farming.**

***Index Terms*—precision agriculture, sustainable agriculture, irrigation system, water conservation, agricultural technology, climate change, remote sensing, machine learning, IoT integration**

I. Introduction

Farmers today face a multitude of challenges that hinder agricultural productivity and sustainability. One of the key issues is the impact of climate change, which has altered weather patterns, reduced water availability, and affected crop yields [1]. The adoption of digital technologies in agriculture—such as IoT, machine learning, and precision farming—offers potential solutions to these challenges, but many farmers still struggle with the high costs and technical expertise required [2]. Furthermore, the shift towards more sustainable farming practices faces resistance due to economic constraints and the difficulty of implementing new techniques at scale [3].

Industry 4.0 has brought significant changes to agriculture by integrating advanced technologies such as the Internet of Things (IoT), artificial intelligence (AI), robotics, and big data analytics into farming practices. These technologies enable precision farming, allowing farmers to optimize crop yields, reduce resource consumption, and increase sustainability. For example, IoT sensors are used to monitor soil moisture and temperature, enabling real-time, data-driven decisions regarding irrigation and fertilization [2]. AI and machine learning models help analyze large datasets to predict crop diseases and pests, allowing farmers to take preventative measures before significant damage occurs [1]. Platforms like Climate FieldView provide farmers with access to real-time data on weather patterns, soil health, and crop conditions, further enhancing decision-making capabilities [2]. These advancements have made agriculture more efficient, data-driven, and sustainable, marking a shift toward smarter, more connected farming practices.

Although technological advancements have transformed many industries, irrigation practices in agriculture still face significant barriers to modernization. Most existing solutions are tailored for large-scale operations and rely heavily on costly technologies, which are unaffordable and impractical for small-scale farmers [1], [2]. For instance, systems like Netafim Smart Irrigation Solutions use advanced drip irrigation technology and remote monitoring to optimize water use. However, their high installation costs and technical complexity make them inaccessible to many farmers. Similarly, RainMachine leverages weather data for irrigation scheduling but fails to incorporate real-time soil conditions and crop-specific requirements, limiting its precision. Another example, FarmBot, employs robotics for automated farming but remains prohibitively expensive and complex, restricting its use to niche markets.

In response, Sustania provides an affordable solution by integrating IoT sensors and machine learning to deliver precise, data-driven irrigation recommendations tailored to local soil, crop, and climate conditions, optimizing water use and enhancing crop productivity [3]. Its user-friendly interface ensures accessibility for farmers with limited technical expertise while also making it a feasible and cost-effective option [4]. This combination of precision, simplicity, and affordability positions Sustania as a viable solution to improve water efficiency and promote sustainable farming practices in an ever-changing world [4].

The Sustania system is designed with several key components that work together to optimize irrigation. It begins with

IoT sensors such as the DHT22 temperature and humidity sensor and a soil moisture sensor, which collect real-time data on temperature, humidity, and soil moisture—factors crucial for making accurate irrigation decisions. This data is sent to the Arduino Nano Controller, which acts as the central hub, collecting inputs from the sensors and forwarding them for analysis. The Data Processing Unit then organizes and cleans the raw data, preparing it for the Machine Learning (ML) Pipeline, where a Random Forest algorithm predicts irrigation needs based on historical and current data. The Decision Engine translates these predictions into actionable irrigation recommendations. Users can view these recommendations and monitor system performance through an interactive user interface dashboard, which also allows manual adjustments if necessary. In addition, System Alerts and Notifications keep users informed about the system status, any issues that arise, and recommendations for immediate action, ensuring that the irrigation process remains efficient and easy to manage.

The objectives of this paper are as follows:

- To design a modular and scalable IoT-enabled irrigation system tailored to small- and medium-scale farming needs.
- To compare Sustania's Random Forest model with existing weather-dependent and robotics-based systems, highlighting its superior performance in cost and functionality.
- To assess the potential of Sustania to address water conservation, improve crop yields, and promote sustainable agricultural practices.

II. Literature Review

Recently, significant efforts have been made to modernize agriculture by exploring how technologies such as IoT sensors, machine learning, and artificial intelligence can ease the workload for farmers and make their jobs less labour-intensive. Several foundational studies on precision agriculture have outlined both the potential and the limitations of this approach. Multiple survey papers provide a general overview of automated irrigation systems, emphasizing the need to move beyond manual and scheduled irrigation toward more intelligent systems. While these papers stress the importance of reducing agricultural water usage, they do not delve into specific technological challenges or solutions [6].

Pandey and Agarwal highlighted the importance of intelligent software solutions for managing irrigation. They recognized the general need to shift away from manual and scheduled irrigation practices in favor of smarter systems. Their work specifically underscored the role of software in optimizing water usage and crop production. However, their research focused primarily on software and did not address the practical integration of this software with physical sensors and actuators—an essential component for real-world implementation. This includes considerations such as sensor selection, data acquisition methods, and communication protocols necessary for interacting with the hardware components of an irrigation system [7].

Building upon this limitation, Khan conducted extensive research on the concept of precision agriculture, addressing the integration of IoT and smart technologies in irrigation. He explored the use of sensors, communication networks, and data analysis. However, the research encountered significant challenges, particularly the high cost of technology and the complexity involved in effectively integrating diverse data sources [8].

The next major advancement in smart agriculture was the incorporation of sustainability as a core objective. In addition to automating farming processes, the focus shifted toward preventing resource wastage, enhancing food security, and minimizing environmental impact. One critical challenge identified was the inability of many systems to provide real-time updates, often resulting in suboptimal outcomes. Moreover, the integration and management of vast volumes of data, coupled with high infrastructure costs, remained major obstacles. The need for energy-efficient and environmentally sustainable IoT devices also became increasingly urgent [9].

To address the financial barriers to adopting these technologies, researchers have proposed low-cost solutions for irrigation pivots and developed affordable sensors to improve accessibility. However, these systems must be compatible with various irrigation methods and crop types, adding to their complexity. Ensuring the reliability, durability, and accuracy of these cost-effective sensors also remains a significant challenge [10], [11].

The reviewed literature underscores the remarkable progress made in the pursuit of smart and sustainable agricultural practices. These pioneering works have established a strong conceptual and technological foundation, offering invaluable insights that have guided our research journey. While these studies have laid the groundwork, they have also revealed limitations and challenges that warrant further exploration. Inspired by their contributions, we sought to address these gaps by refining methodologies, optimizing models, and integrating cost-effective, practical solutions tailored to diverse agricultural contexts.

Our research focuses on optimizing water usage while maintaining crop yield through the integration of a low-cost moisture sensor [7]. Additionally, we have incorporated a temperature sensor into our system. Both sensors are designed to provide real-time data, which are used to make predictions based on the collected information. We employed the Random Forest Classifier as the machine learning model for training our data, drawing inspiration from the works of Priyanka, Joseph, and Naik. Their research explored various machine learning models to analyze data from smart sensors for predicting crop stages, irrigation schedules, and fertilizer requirements. However, their studies did not validate or optimize the models for application across diverse crops and farming practices [12]. In our study, we successfully optimized the Random Forest Classifier to support multiple crop patterns. Nonetheless, significant opportunities remain for further research and improvement in this field.

Our contributions extend beyond incremental improve-

ments; they aim to enhance the technological framework of smart agriculture while prioritizing sustainability and accessibility. By addressing key challenges such as scalability, resource efficiency, and adaptability, we have contributed meaningfully to the existing body of knowledge and advanced the practical application of these technologies. Furthermore, we have endeavored to pave a clear path for future research by offering a robust platform for exploring innovative strategies that address the evolving needs of global agriculture. In doing so, we hope to inspire continued progress in this vital field, bridging the gap between emerging technologies and the practical realities faced by farmers worldwide.

III. Proposed Methodology

This section presents a comprehensive breakdown of the key stages involved in implementing our proposed smart irrigation system. We begin with a high-level overview, including a diagrammatic representation of the system's core architecture, to help readers visualize how the various components interact. This initial subsection also explains the primary functionalities that enable the system to operate efficiently and intelligently.

Subsequent subsections delve deeper into the technical aspects, beginning with the integration of Internet of Things (IoT) devices such as temperature, humidity, and soil moisture sensors. These devices form the backbone of the system by collecting real-time environmental data. We describe the methods used to interface these sensors with the microcontroller and how the collected data is processed for further use.

Following this, we provide a comparative evaluation of several machine learning models that were tested to determine the most effective approach for predicting irrigation needs. Each model is assessed based on performance metrics such as accuracy, precision, recall, and cross-validation efficiency.

Additionally, we discuss in detail the dataset used to train these models, including how it was obtained, preprocessed, and expanded to suit the needs of our application. This dataset forms the foundation of the system's intelligence, enabling it to make informed irrigation decisions. Its structure, diversity, and relevance to real-world agricultural conditions are emphasized to highlight its critical role in system development and performance.

A. System Architecture

Sustania's architecture is designed for scalability and user-focused efficiency, integrating the key modules as shown in the diagram.

Fig. 1 illustrates a smart irrigation system that uses sensors (for temperature, humidity, and soil moisture), weather data, and an Arduino Nano controller to collect and process data. A machine learning pipeline (Random Forest) analyzes the data to generate irrigation recommendations via a decision engine. The system integrates a user dashboard for monitoring, manual controls, and reporting, while storing data in a database for analytics and performance tracking. It ensures efficient water usage through real-time alerts and actionable insights.

The system architecture integrates several key components to enable efficient and data-driven irrigation management. Sensors, including the DHT22 temperature and humidity sensor and a soil moisture sensor, collect real-time environmental data, complemented by Weather API inputs for forecasts and rainfall trends. An Arduino Nano microcontroller acts as the central hub, collecting raw sensor data and forwarding it to the Data Processing Unit, where the data is cleaned and organized for analysis. A machine learning pipeline, using the Random Forest algorithm, predicts irrigation needs based on historical and real-time inputs, while the Decision Engine translates these predictions into actionable recommendations, such as irrigation schedules and quantities [12]. The user interface dashboard provides real-time metrics, analytics, and manual parameter adjustment options, ensuring transparency and control. Additionally, a notification system delivers timely alerts and updates about system performance and environmental conditions, enhancing user awareness and system reliability.

B. IoT Integration

Our system utilizes a resistive soil moisture sensor alongside a temperature and humidity sensor to provide real-time data and generate predictions. In the initial stages of the project, the sensors were configured on a Raspberry Pi. However, due to the absence of an integrated analog-to-digital converter required to process analog readings from the soil moisture sensor, we transitioned to an Arduino Nano controller.

For the temperature and humidity sensor, we initially used the DHT11 model. Through repeated testing and evaluation, we identified the DHT22 sensor as a more durable and reliable alternative, leading to its adoption in place of the DHT11. This adjustment improved the system's overall performance and longevity [13].

C. Model Comparison and Training

a) : The initial dataset for our project, sourced from Kaggle, contained data on moisture and temperature levels for cotton plants, along with predictions on whether irrigation was required. To adapt this dataset for our study, we developed a Python script to modify and scale the temperature and irrigation values to align with the requirements of wheat crops. Additionally, we expanded the original dataset, which consisted of approximately 500 entries, to 5,000 entries, creating a new semi-synthesized dataset. This process ensured a more comprehensive dataset for training and validating our system.

This newly generated dataset was first preprocessed and then used to train multiple machine learning models. When these models were initially run in parallel, they all showed 100% accuracy—indicating signs of overfitting. To address this, bias was intentionally introduced to reduce variance and mitigate overfitting. The modified dataset was then used to retrain all the machine learning models. Among these, the Random Forest model emerged as the most effective, achieving the highest accuracy while maintaining optimal cross-validation time. Thus all the evaluations we have conducted justifies the use of Random Forest model [14], [15].

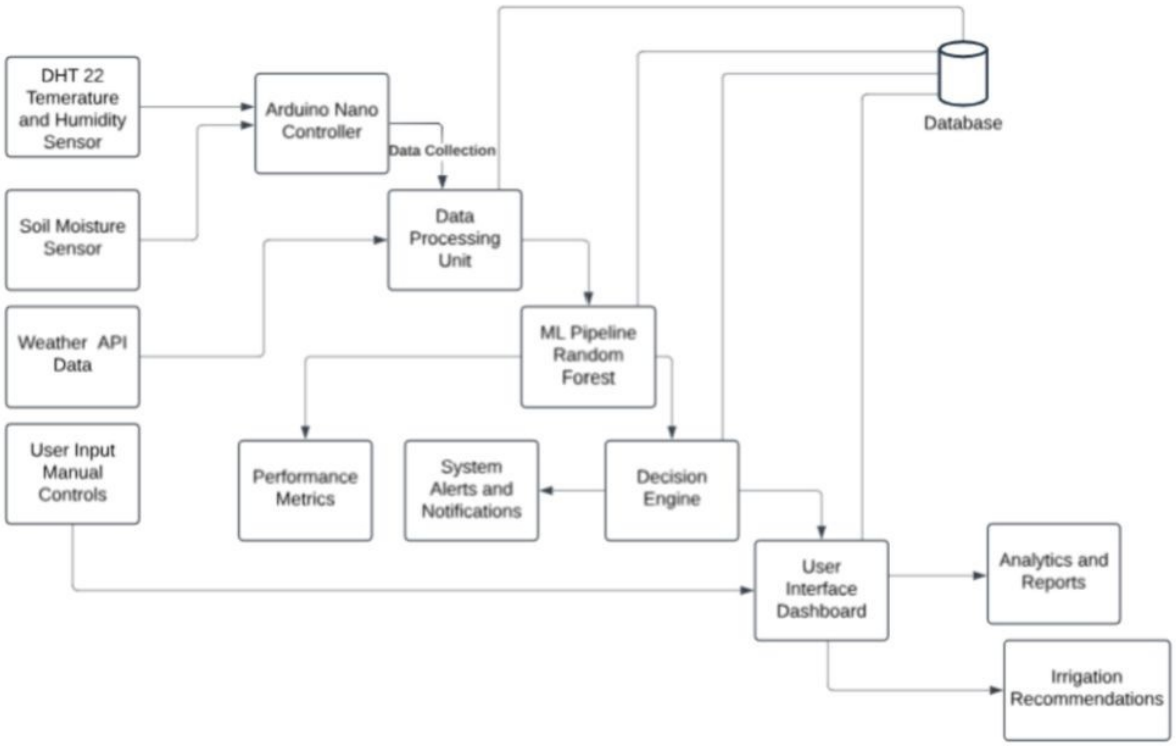


Fig. 1. System Architecture Diagram

IV. Evaluation and Analysis

In this section, we present the results obtained and provide a detailed analysis of the system's performance. The subsection "Results" contains a screenshot of the main Farmer Dashboard and a brief description of the information displayed. The subsection "Performance Analysis" presents the ROC curves and graphs for the different machine learning models, as well as the confusion matrix for the selected model.

A. Results

The system generates detailed performance metrics and analytical reports, offering insights into environmental conditions and overall system efficiency. Based on these insights, it provides irrigation recommendations, helping users implement precise and optimal watering strategies tailored to the specific needs of their crops or plants.

Fig. 2 shows the real-time implementation of our irrigation prediction system, where the moisture and temperature sensors connected to the Arduino Nano controller provide real-time updates on the Farmer Dashboard along with irrigation predictions based on these readings.

Fig. 3 shows the Farmer Dashboard page, which provides real-time readings of soil moisture and temperature along with irrigation predictions.



Fig. 2. Real-Time Implementation of Irrigation Prediction System

B. Performance Analysis

The machine learning model applied in Sustania proactively predicts irrigation needs with high accuracy. According to the evaluation metrics, the system achieves an accuracy of 93.94%, indicating that almost all predictions made by the model are correct. Furthermore, the precision score of 1.0000 means that every predicted irrigation event was indeed necessary, thereby eliminating unnecessary water usage. The recall score of 0.8788 indicates that most instances requiring irrigation were correctly identified, ensuring minimal risk of under-irrigation. These results demonstrate the robustness of the

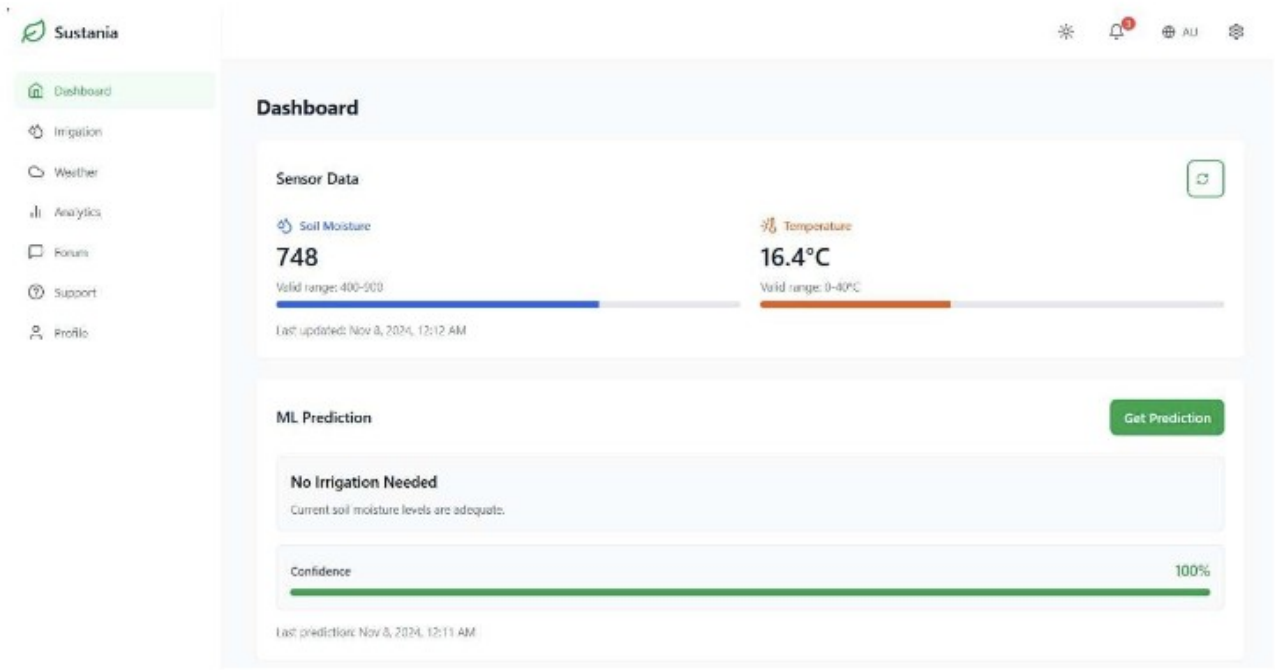


Fig. 3. Sensor Readings along with the Predicted Outcome

Random Forest algorithm in handling diverse environmental inputs for accurate prediction.

Fig. 4 shows a comparison of performance metrics—accuracy, F1 score, precision, and recall—for several machine learning models. By laying out these details, it helps in identifying which model is the most effective for the task, making it easier to understand the trade-offs and choose the best approach.

Fig. 5 shows the ROC curve for different ML Models. The figure clearly shows why Random Forest was selected for our system. AUC for Random Forest is one of the highest among all the models we had tested. Though Gradient Boosting too has the same AUC, Random Forest was selected because it is a little faster than Gradient Boosting.

TABLE I
SIMPLIFIED CLASSIFICATION REPORT

Metric	Precision	Recall	F1-Score
Accuracy		0.91	
Macro Average	0.91	0.91	0.91
Weighted Average	0.91	0.91	0.91

Table I shows the performance assessment of the trained ML model. A class-wise breakdown of precision, recall, and F1-score was computed, but only summary metrics are shown here for clarity. The output shows a machine learning model trained with 91% accuracy. The best parameters were tuned, and the model was saved as irrigation model.joblib.

Figure 6 shows the confusion matrix of the Random Forest model selected for training. The confusion matrix visualizes the performance of the Random Forest classifier by displaying the counts of true positives, true negatives, false positives, and false negatives. It provides a comprehensive summary of the classifier's accuracy and error distribution across predicted and actual classes.

Sustania is modular and flexible, making it easily adaptable to diverse farming conditions and scales. Its design supports the integration of additional sensors—such as those measuring soil nutrients or weather conditions—making it suitable for more complex farming operations. Whether on small-scale farms or large agricultural estates, Sustania's scalability ensures compatibility across various farm sizes and crop types. This provides a significant advantage in terms of widespread adoption, as it can be tailored to the specific needs of different farming communities.

Moreover, the system's affordability and ease of use make it accessible to small and medium-scale farmers, who often cannot afford high-end technological solutions. Due to its efficiency in operating under a variety of agricultural conditions, Sustania can be considered a truly versatile solution—especially in regions where water is scarce or resources are constrained. These characteristics highlight Sustania as a sustainable and viable tool for modern agriculture, offering better water management and improved productivity on a large scale.

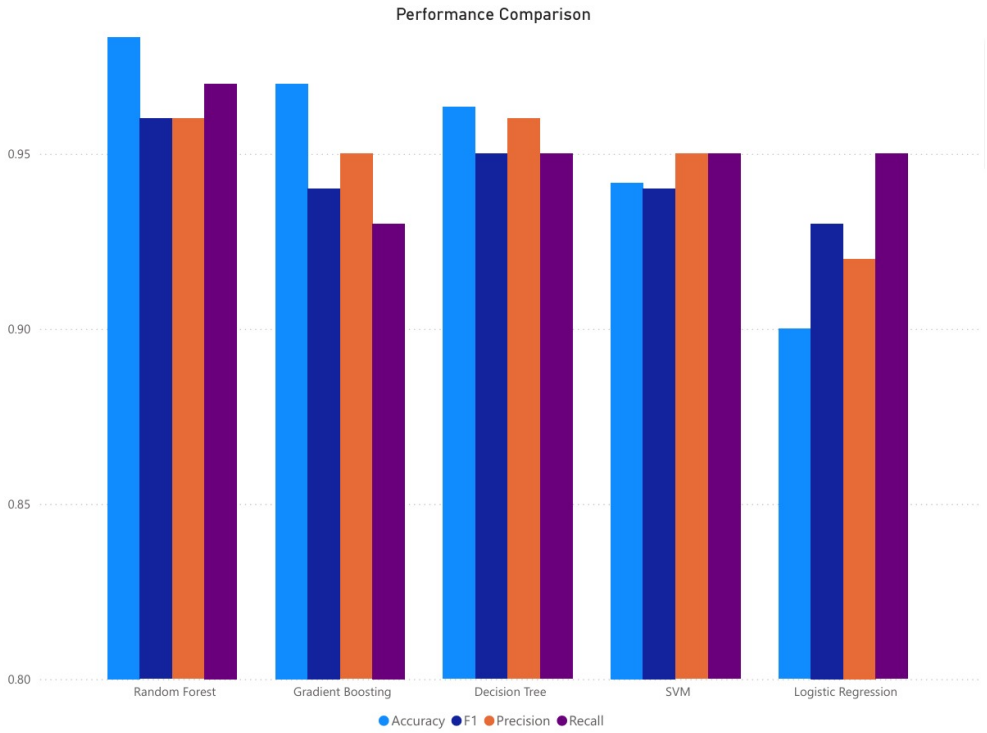


Fig. 4. Performance Comparison Graph

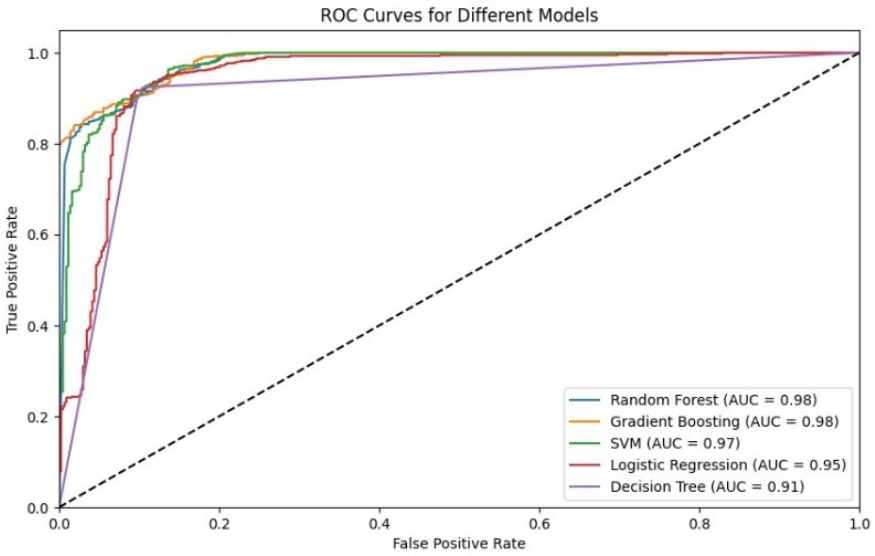


Fig. 5. ROC Curve

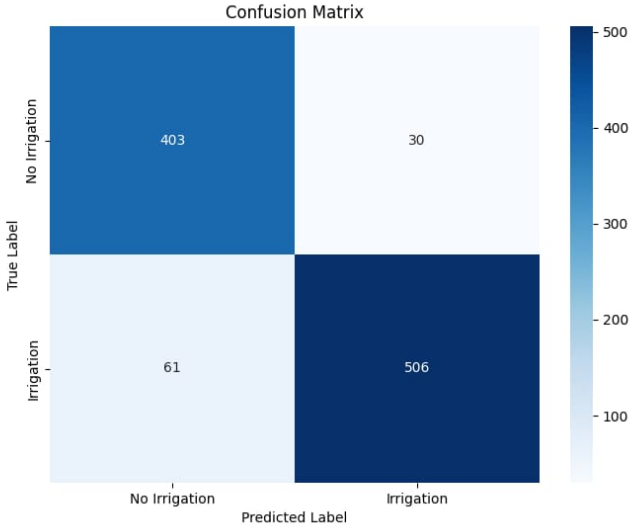


Fig. 6. Confusion Matrix

V. Conclusion and Future Scope

Sustania represents a key advancement in the evolution of precision farming. It combines the Internet of Things, machine learning, and an interactive interface to tackle irrigation challenges faced by small- to medium-scale farmers. By optimizing water usage and minimizing waste, it establishes itself as both a sustainable and cost-effective solution. With these two principles at its core, Sustania bridges the gap between advanced agricultural technology and resource-constrained farming communities, promoting widespread adoption of smart irrigation systems.

One of the challenges encountered during the system's implementation was its adaptability to varying environmental and regional conditions. This can be addressed by using localized training data to improve the accuracy of machine learning models. Partnering with agricultural experts to fine-tune the system for specific crop needs and soil profiles offers a more long-term and stable solution.

Scalability also posed a concern: how the system could adapt to different farm sizes and crop types, and how to integrate additional sensors without increasing complexity. The answer lies in a modular design that enables incremental hardware upgrades for added functionality. Training machine learning models on datasets representing diverse conditions will further enhance the system's adaptability.

Despite these challenges, Sustania holds great potential for growth and innovation. Advanced sensors—such as those detecting soil nutrient levels—could be added to enhance crop management beyond water usage. Long-term irrigation needs could be forecasted using AI by analyzing historical data, weather patterns, and crop cycles, helping farmers plan better and ensure long-term sustainability in water management.

A dedicated mobile application would further increase accessibility by providing farmers with instant updates, control options, and relevant insights directly on their smartphones. These enhancements would not only improve user experience but also extend the system's reach to remote agricultural communities.

Moreover, partnerships with governmental and non-governmental organizations could accelerate widespread adoption by providing financial support or subsidies for farmers. Collaborations with agricultural research institutions could further enrich the platform with region-specific models tailored to local needs. These initiatives would promote the adoption of smart irrigation systems and equip farmers with customized tools and expertise to improve productivity and sustainability.

Sustania has the potential to become a cornerstone in the innovation of precision agriculture by fostering supportive networks of innovation and collaboration that drive both economic growth and environmental stewardship.

These future developments will reinforce Sustania's role in promoting sustainable farming practices, helping to address global challenges such as water scarcity, food security, and climate change. By continuously evolving and incorporating new technologies, Sustania can redefine irrigation management and support the long-term growth of agriculture worldwide.

References

[1] M. L. T. Dixon and P. P. Srinivasan, "Climate change impacts on agriculture: Implications for food security and farm sustainability," *Agricultural Systems*, vol. 180, pp. 85-96, 2021.

[2] R. C. S. Kumar and N. S. P. R. Krishna, "Agriculture 4.0: The future of farming technology," *Food Science & Technology International*, vol. 28, no. 1, pp. 15-30, 2022.

[3] K. S. Ghosh and S. K. Agarwal, "Sustainable farming practices for the 21st century: Challenges and future prospects," *Renewable Agriculture and Food Systems*, vol. 37, no. 2, pp. 170-185, 2022.

[4] L. G. Morales and H. V. Turner, "Labor shortages in agriculture: Causes and solutions," *Journal of Agricultural Economics*, vol. 49, no. 3, pp. 320-334, 2023.

[5] Jain, S.; Vani, K.S. A survey of the automated irrigation systems and the proposal to make the irrigation system intelligent. Int. J. Comput. Sci. Eng. 2018, 6, 357–360.

[6] P. Pandey and S. Agarwal, "A low-cost smart irrigation planning based on machine learning and Internet of Things," *Current Agriculture Research Journal*, ISSN: 2347-4688, Vol. 11, No.(2) 2023, pg. 568-579.

[7] Khan Z., "The integration of Internet of Things (IoT) in precision agriculture," AGBIR.2024;40(4):1194-1197.

[8] Dewan Md Nur Anjum Ashir, Md. Taimur Ahad, Manosh Talukder, Tahsinur Rahman, "Internet of Things (IoT) based Smart Agriculture Aiming to Achieve Sustainable Goals," arXiv:2206.06300 [cs.CY].

[9] O. Debauche, M. El Moulat, S. Mahmoudi, P. Manneback and F. Lebeau, "Irrigation pivot-center connected at low cost for the reduction of crop water requirements," *2018 International Conference on Advanced Communication Technologies and Networking (CommNet)*, Marrakech, Morocco, 2018, pp. 1-9, doi: 10.1109/COMMNET.2018.8360259.

[10] Sandra Sendra, Lorena Parra, Vicente Ortuño, Jaime Lloret, "A Low Cost Turbidity Sensor Development," *SENSORCOMM 2013 : The Seventh International Conference on Sensor Technologies and Applications*, ISBN: 978-1-61208-296-7.

[11] B. Priyanka. (2024), "Machine Learning Driven Precision Agriculture: Enhancing Farm Management through Predictive Insights," *International Journal of Intelligent Systems and Applications in Engineering*, 12(23s), 195–201.

[12] Muthuramalingam R, Rathnam Velu R, Baskar H, Vellan Saminathan MH, "An IoT-Based Smart Irrigation System," Engineering Proceedings, 2024;66(1):13.doi:10.3390/engproc2024066013.

[13] García L, Parra L, Jimenez JM, Lloret J, Lorenz P, "IoT-Based Smart Irrigation Systems: An Overview on the Recent Trends on Sensors and IoT Systems for Irrigation in Precision Agriculture," Sensors (Basel). 2020 Feb 14;20(4):1042. doi: 10.3390/s20041042. PMID: 32075172; PMCID: PMC7070544.

[14] A. Yahia, W. Menasri, D. Cherifi and A. Gacemi, "Performance Evaluation of Machine Learning Algorithms for Smart Irrigation Systems," *2023 2nd International Conference on Electronics, Energy and Measurement (IC2EM)*, Medea, Algeria, 2023, pp. 1-5, doi: 10.1109/IC2EM59347.2023.10419485.

[15] Kim, Hak Jin et al, "Soil macronutrient sensing for precision agriculture," *Journal of environmental monitoring*, JEM 11 10 (2009): 1810-24 .